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If the smaller of 2 consecutive odd integers is a multiple of 5, which of the following could NOT be the sum of these 2 integers?

A. –8

B. 12

C. 22

D. 52

E. 252

Algebra Operations with Integers

Since the smaller of the 2 consecutive odd integers is a multiple of 5, let it be represented by $5n$ for some integer n. Then the other odd integer can be represented by $5n+2$. The sum of these two integers is $10n+2$. The sum is –8 when $n=-1$ and $5n=(5)(-1)$ is odd. The sum is 12 when $n=1$ and $5n=(5)(1)$ is odd. The sum is 22 when $n=2$, but $5n=(5)(2)$ is not odd. There is no need to check the D and E because it has been determined that 22 cannot be the sum of the 2 consecutive odd integers. For completeness, the sum is 52 when $n=5$ and $5n=(5)(5)$ is odd. The sum is 252 when $n=25$ and $5n=(5)(25)$ is odd.

The correct answer is C.

•

Eight light bulbs numbered 1 through 8 are arranged in a circle as shown above. The bulbs are wired so that every third bulb, counting in a clockwise direction, flashes until all bulbs have flashed once. If the bulb numbered 1 flashes first, which numbered bulb will flash last?

A. 2

B. 3

C. 4

D. 6

E. 7

Arithmetic Properties of Integers

The easiest way to do this problem might be by just counting every third bulb going clockwise around the circle starting at Bulb 1, which flashes, skipping 2 bulbs and getting to Bulb 4, which flashes, skipping 2 bulbs and getting to Bulb 7, which flashes, skipping 2 bulbs and getting to Bulb 2, which flashes, skipping 2 bulbs and getting to Bulb 5, which flashes, skipping 2 bulbs and getting to Bulb 8, which flashes, skipping 2 bulbs and getting to Bulb 3, which flashes, and finally skipping 2 bulbs and getting to Bulb 6, which flashes. Now, all 8 bulbs have flashed once and the last one to flash was Bulb 6.

The correct answer is D.

Closing Prices of Stock X
During a Certain Week
(in dollars)

Monday	Tuesday	Wednesday	Thursday	Friday

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21	19	22	$24\frac{1}{2}$	23

•

A certain financial analyst defines the "volatility" of a stock during a given week to be the result of the following procedure: find the absolute value of the difference in the stock's closing price for each pair of consecutive days in the week and then find the average (arithmetic mean) of these 4 values. What is the volatility of Stock X during the week shown in the table?

A. 0.50

B. 1.80

C. 2.00

D. 2.25

E. 2.50

Arithmetic Statistics

The volatility of Stock X during the week is the average of the 4 values associated with the 4 pairs of consecutive days during the week.

$$\frac{(\text{Mon \& Tue}) + (\text{Tue \& Wed}) + (\text{Wed \& Thu}) + (\text{Thu \& Fri})}{4}$$

$$= \frac{|19-21| + |22-19| + |24.5-22| + |23-24.5|}{4}$$

$$= \frac{|-2| + |3| + |2.5| + |-1.5|}{4} = \frac{2+3+2.5+1.5}{4} = \frac{9}{4} = 2.25$$

The correct answer is D.

•

If $y = \frac{|3x-5|}{-x^2-3}$, for what value of x will the value of y be greatest?

A. –5

B. $-\frac{3}{5}$

C. 0

D. $\frac{3}{5}$

E. $\frac{5}{3}$

Algebra Functions; Absolute Value

Since the absolute value of any real number is greater than or equal to zero, it follows that $|3x-5| \geq 0$. Also, for any real number x we have $x^2 \geq 0$, and hence $-x^2 \leq 0$. Subtracting 3 from both sides of the last inequality gives $-x^2 - 3 \leq -3$. Therefore, the numerator of the expression for y is greater than or equal to zero and the denominator of the expression for y is negative. It follows that the value of y cannot be greater than 0. However, the value of y is equal to 0 when $|3x-5| = 0$, or $3x-5=0$, or $x = \frac{5}{3}$. Therefore, the value of x for which the value of y is greatest (i.e., when $y=0$) is $x = \frac{5}{3}$.

The correct answer is E.

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•

What values of x have a corresponding value of y that satisfies both $xy > 0$ and $xy = x + y$?

A. $x \leq -1$

B. $-1 < x \leq 0$

C. $0 < x \leq 1$

D. $x > 1$

E. All real numbers

Algebra Equations; Inequalities

First, use $xy = x + y$ to solve for y in terms of x.

$$\begin{aligned} xy &= x + y && \text{given} \\ xy - y &= x && \text{subtract } y \text{ from both sides} \\ y(x-1) &= x && \text{factor} \\ y &= \frac{x}{x-1} && \text{divide both sides by } x-1 \end{aligned}$$

Note that the division by $x - 1$ requires $x \neq 1$, and thus $x = 1$ needs to be considered separately. However, if $x = 1$, then $xy = x + y$ becomes $y = 1 + y$, which is not true for any value of y. Using $y = \frac{x}{x-1}$, it follows that the inequality $xy > 0$ is equivalent to $\frac{x^2}{x-1} > 0$. Since $x^2 \geq 0$ for each value of x, the quotient $\frac{x^2}{x-1}$ can only be positive when $x \neq 0$ and $x - 1$ is positive, or when $x > 1$.

Alternatively, the correct answer can be found by eliminating the incorrect answers, which can be accomplished by considering the endpoints of the intervals given in the answer choices.

Case 1: If $x = -1$, then $xy = x + y$ becomes $-y = -1 + y$, or $y = \frac{1}{2}$. However, in this case $xy = (-1)\left(\frac{1}{2}\right)$ is negative, and thus $xy > 0$ is not true. Therefore, the answer cannot be A or E.

Case 2: If $x = 0$, then $xy = 0$, and thus $xy > 0$ is not true. Therefore, the answer cannot be B or E.

Case 3: If $x = 1$, then $xy = x + y$ becomes $y = 1 + y$, which is not true for any value of y. Therefore, the answer cannot be C or E.

Since the answer cannot be A, B, C, or E, it follows that the answer is D.

The correct answer is D.

•

Employee X's annual salary is $12,000 more than half of Employee Y's annual salary. Employee Z's annual salary is $15,000 more than half of Employee X's annual salary. If Employee X's annual salary is $27,500, which of the following lists these three people in order of increasing annual salary?

A. Y, Z, X

B. Y, X, Z

C. Z, X, Y

D. X, Y, Z

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D. X, Y, Z

E. X, Z, Y

Algebra First-Degree Equations

Letting x, y, and z represent the annual salary, in dollars, of Employee X, Employee Y, and Employee Z, respectively, the following information is given:

1. $x = 12{,}000 + \frac{y}{2}$
2. $x = 15{,}000 + \frac{x}{2}$
3. $x = 27{,}500$

From (1) and (3), it follows that $27{,}500 = 12{,}000 + \frac{y}{2}$ or $y = 2(27{,}500 - 12{,}000) = 31{,}000$. From (2) and (3), it follows that $z = 15{,}000 + \frac{27{,}500}{2} = 28{,}750$. Therefore, $x < z < y$.

The correct answer is E.

•

$$C = \begin{cases} 0.10s, \text{ if } s \leq 60{,}000 \\ 0.10s + 0.04(s - 60{,}000), \text{ if } s > 60{,}000 \end{cases}$$

The formula above gives the contribution C, in dollars, to a certain profit-sharing plan for a participant with a salary of s dollars. How many more dollars is the contribution for a participant with a salary of $70,000 than for a participant with a salary of $50,000?

A. $800

B. $1,400

C. $2,000

D. $2,400

E. $2,800

Algebra Applied Problems; Formulas

For a participant with a salary of $70,000, $C = 0.1(\$70{,}000) + 0.04(\$70{,}000 - \$60{,}000) = \$7{,}000 + \$400 = \$7{,}400$. For a participant with a salary of $\$50{,}000$, $C = 0.1(\$50{,}000) = \$5{,}000$. The difference is $\$7{,}400 - \$5{,}000 = \$2{,}400$.

The correct answer is D.

•

Next month, Ron and Cathy will each begin working part-time at $\frac{3}{5}$ of their respective current salaries. If the sum of their reduced salaries will be equal to Cathy's current salary, then Ron's current salary is what fraction of Cathy's current salary?

A. $\frac{1}{3}$

B. $\frac{2}{5}$

C. $\frac{1}{2}$

D. $\frac{3}{5}$

E. $\frac{2}{}$

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E. $\frac{}{3}$

Algebra First-Degree Equations

Letting R and C, respectively, represent Ron's and Cathy's current salaries, it is given that $\frac{3}{5}R + \frac{3}{5}C = C$. It follows that $\frac{3}{5}R = \frac{2}{5}C$ and $R = \frac{5}{3}\left(\frac{2}{5}C\right) = \frac{2}{3}C$.

The correct answer is E.

•

David and Ron are ordering food for a business lunch. David thinks that there should be twice as many sandwiches as there are pastries, but Ron thinks the number of pastries should be 12 more than one-fourth of the number of sandwiches. How many sandwiches should be ordered so that David and Ron can agree on the number of pastries to order?

A. 12

B. 16

C. 20

D. 24

E. 48

Algebra Simultaneous Equations

Let S be the number of sandwiches that should be ordered and let P be the number of pastries that should be ordered. Then David desires $S = 2P$ and Ron desires $P = 12 + \frac{1}{4}S$.

$S = 2P$	given
$S = 2(12 + \frac{1}{4}S)$	$P = 12 + \frac{1}{4}S$
$S = 24 + \frac{1}{2}S$	distributive law
$\frac{1}{2}S = 24$	subtract $\frac{1}{2}S$ from both sides
$S = 48$	multiply both sides by 2

The correct answer is E.

•

The cost of purchasing each box of candy from a certain mail order catalog is v dollars per pound of candy, plus a shipping charge of h dollars. How many dollars does it cost to purchase 2 boxes of candy, one containing s pounds of candy and the other containing t pounds of candy, from this catalog?

A. $h + stv$

B. $2h + stv$

C. $2hstv$

D. $2h + s + t + v$

E. $2h + v(s + t)$

Algebra Formulas

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The cost, in dollars, to purchase the 2 boxes of candy is the sum of 2 shipping charges and the cost of $s + t$ pounds of candy.

$$\begin{aligned} \text{cost} &= (2 \text{ shipping charges}) + (v)(s+t) \\ \text{cost} &= 2(h) + (v)(s+t) \\ \text{cost} &= 2h + v(s+t) \end{aligned}$$

The correct answer is E.

•

If $x \neq -\frac{1}{2}$, then $\frac{6x^3 + 3x^2 - 8x - 4}{2x+1} =$

A. $3x^2 + \frac{3}{2}x - 8$

B. $3x^2 + \frac{3}{2}x - 4$

C. $3x^2 - 4$

D. $3x - 4$

E. $3x + 4$

Algebra Factoring

$$\begin{aligned} \frac{6x^3 + 3x^2 - 8x - 4}{2x+1} &= \frac{(6x^3 + 3x^2) - (8x + 4)}{2x+1} && \text{group} \\ &= \frac{3x^2(2x+1) - 4(2x+1)}{2x+1} && \text{factor} \\ &= \frac{(3x^2 - 4)(2x+1)}{2x+1} && \text{factor} \\ &= 3x^2 - 4 && \text{cancel since } x \neq \frac{1}{2} \end{aligned}$$

Alternatively, sometimes it is easier or quicker to test one-variable expressions for equality by substituting a convenient value for the variable and eliminating answer choices for which the value of the expression in that answer choice does not equal the value of the given expression. For example, choose $x = 0$, since calculations for $x = 0$ are minimal. Then, as shown in the table below, $\frac{6x^3 + 3x^2 - 8x - 4}{2x+1} = \frac{-4}{1} = -4$, but $3x^2 + \frac{3}{2}x - 8 = -8$ and $3x + 4 = 4$, neither of which equals –4, so answer choices A and E can be eliminated. Another convenient value to choose for x is 1. There is no need to evaluate answer choices A and E at 1 since they have already been eliminated. As shown, when $x = 1$, $\frac{6x^3 + 3x^2 - 8x - 4}{2x+1} = -1$, but $3x^2 + \frac{3}{2}x - 4 = \frac{1}{2} \neq -1$, so answer choice B can be eliminated. A third convenient value for x is –1. There is no need to evaluate answer choices A, B, and E at –1 since they have already been eliminated. As shown, when $x = -1$, $\frac{6x^3 + 3x^2 - 8x - 4}{2x+1} = -1$, but $3x - 4 = -7 \neq -1$, so answer choice D can be eliminated. Note that, if $x = -1$ had been chosen initially, A, B, D, and E would have been eliminated immediately since $3x^2 + \frac{3}{2}x - 8 = -6\frac{1}{2} \neq -1$, $3x^2 + \frac{3}{2}x - 4 = -2\frac{1}{2} \neq -1$, $3x - 4 = -7 \neq -1$, and $3x + 4 = 1 \neq -1$.

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		$x = 0$	$x = 1$	$x = -1$
	$\frac{6x^3 + 3x^2 - 8x - 4}{2x + 1}$	–4	–1	–1
A	$3x^2 + \frac{3}{2}x - 8$	–8		
B	$3x^2 + \frac{3}{2}x - 4$	–4	$\frac{1}{2}$	
C	$3x^2 - 4$	–4	–1	–1
D	$3x - 4$	–4	–1	–7
E	$3x + 4$	4		

The correct answer is C.

•

If $x^2 + bx + 5 = (x + c)^2$ for all numbers x, where b and c are positive constants, what is the value of b?

A. $\sqrt{5}$

B. $\sqrt{10}$

C. $2\sqrt{5}$

D. $2\sqrt{10}$

E. 10

Algebra Second-Degree Equations

Given that $x^2 + bx + 5 = (x + c)^2$, since $(x + c)^2 = x^2 + 2cx + c^2$, it follows that $5 = c^2$ and $b = 2c$. The possible values of c are $-\sqrt{5}$ and $\sqrt{5}$, but since c is positive, $c = \sqrt{5}$ and $b = 2c = 2\sqrt{5}$.

The correct answer is C.

•

Last year Shannon listened to a certain public radio station 10 hours per week and contributed $35 to the station. Of the following, which is closest to Shannon's contribution per minute of listening time last year?

A. $0.001

B. $0.010

C. $0.025

D. $0.058

E. $0.067

Arithmetic Measurement Conversion

Since there are 52 weeks in 1 year and 60 minutes in 1 hour, 10 hours per week is equivalent to (10)(52)(60) = 31,200 minutes per year. Shannon's $35 contribution is then $\frac{35}{31{,}200}$ dollars per minute, which is closest to $0.001 per minute.

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The correct answer is A.

•

Each of the 20 employees at Company J is to receive an end-of-year bonus this year. Agnes will receive a larger bonus than any other employee, but only $500 more than Cheryl will receive. None of the employees will receive a smaller bonus than Cheryl. If the amount of money to be distributed in bonuses at Company J this year totals $60,000, what is the largest bonus Agnes can receive?

A. $3,250

B. $3,325

C. $3,400

D. $3,475

E. $3,500

Algebra Applied Problems

Since the total amount of the bonuses is fixed, the largest possible bonus that Agnes can receive will occur when the total amount received by the 19 employees other than Agnes is the smallest possible. Let A be the bonus, in dollars, that Agnes receives. Then, in dollars, Cheryl will receive $(A - 500)$, and each of the remaining 18 employees will receive between $(A - 500)$ and A. Therefore, the total amount received by the 19 employees other than Agnes is smallest when each of these 19 employees receives $(A - 500)$ dollars.

$$19(A - 500) + A = 60{,}000 \quad \text{total of 20 bonuses is \$60,000}$$

$$19A - 9{,}500 + A = 60{,}000 \quad \text{distributive law}$$

$$20A - 9{,}500 = 60{,}000 \quad \text{combine like terms}$$

$$20A = 69{,}500 \quad \text{add 9,500 to both sides}$$

$$A = 3{,}475 \quad \text{divide both sides by 20}$$

The correct answer is D.

•

Beth, Naomi, and Juan raised a total of $55 for charity. Naomi raised $5 less than Juan, and Juan raised twice as much as Beth. How much did Beth raise?

A. $9

B. $10

C. $12

D. $13

E. $15

Algebra Simultaneous Equations

Let B, N, and J be the amounts raised, respectively and in dollars, by Beth, Naomi, and Juan.

$$B + N + J = 55 \quad \text{given}$$

$$N = J - 5 \quad \text{given}$$

$$J = 2B \quad \text{given}$$

$$J = 55 - B - N \quad \text{subtract } B + N \text{ from both sides of first equation}$$

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$$J = 55 - B - (J - 5) \qquad N = J - 5$$

$$2B = 55 - B - (2B - 5) \qquad J = 2B$$

$$B = 12 \qquad \text{solve for } B$$

The correct answer is C.

•

The set of solutions for the equation $\left(x^2 - 25\right)^2 = x^2 - 10x + 25$ contains how many real numbers?

A. 0
B. 1
C. 2
D. 3
E. 4

Algebra Second-Degree Equations

$$\left(x^2 - 25\right)^2 = x^2 - 10x + 25 \quad \text{given}$$

$$(x+5)^2(x-5)^2 = (x-5)^2 \quad \text{factor}$$

$$(x+5)^2(x-5)^2 - (x-5)^2 = 0 \quad \text{subtract } (x-5)^2$$

$$(x-5)^2[(x+5)^2 - 1] = 0 \quad \text{factor}$$

$$(x-5)^2[(x+5) - 1][(x+5) + 1] = 0 \quad \text{factor}$$

$$(x-5)^2(x+4)(x+6) = 0 \quad \text{subtraction, addition}$$

Thus, the solution set of $\left(x^2 - 25\right)^2 = x^2 - 10x + 25$ contains 3 real numbers: 5, –4, and –6.

The correct answer is D.

•

An aerosol can is designed so that its bursting pressure, B, in pounds per square inch, is 120% of the pressure, F, in pounds per square inch, to which it is initially filled. Which of the following formulas expresses the relationship between B and F?

A. $B = 1.2F$
B. $B = 120F$
C. $B = 1 + 0.2F$
D. $B = \dfrac{F}{1.2}$
E. $B = \dfrac{1.2}{F}$

Algebra Formulas; Percents

We are given that B is 120% of F, so $B = (120\%)F$, or $B = 1.2F$. Note that both B and F are given in pounds per square inch, so there are no unit conversions involved.

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The correct answer is A.

•

The average (arithmetic mean) of the positive integers x, y, and z is 3. If $x < y < z$, what is the greatest possible value of z?

A. 5

B. 6

C. 7

D. 8

E. 9

Algebra Inequalities

It is given that $\frac{x+y+z}{3} = 3$, or $x + y + z = 9$, or $z = 9 + (-x - y)$. It follows that the greatest possible value of z occurs when $-x - y = -(x + y)$ has the greatest possible value, which occurs when $x + y$ has the least possible value. Because x and y are different positive integers, the least possible value of $x + y$ occurs when $x = 1$ and $y = 2$. Therefore, the greatest possible value of z is 9 – 1 – 2 = 6.

The correct answer is B.

•

The product of 3,305 and the 1-digit integer x is a 5-digit integer. The units (ones) digit of the product is 5 and the hundreds digit is y. If A is the set of all possible values of x and B is the set of all possible values of y, then which of the following gives the members of A and B?

	A	B
(A)	{1, 3, 5, 7, 9}	{0, 1, 2, 3, 4, 5, 6, 7, 8, 9}
(B)	{1, 3, 5, 7, 9}	{1, 3, 5, 7, 9}
(C)	{3, 5, 7, 9}	{1, 5, 7, 9}
(D)	{5, 7, 9}	{1, 5, 7}
(E)	{5, 7, 9}	{1, 5, 9}

Arithmetic Properties of Numbers

Since the products of 3,305 and 1, 3,305 and 2, and 3,305 and 3 are the 4-digit integers 3,305, 6,610, and 9,915, respectively, it follows that x must be among the 1-digit integers 4, 5, 6, 7, 8, and 9. Also, since the units digit of the product of 3,305 and x is 5, it follows that x cannot be 4 (product has units digit 0), 6 (product has units digit 0), or 8 (product has units digit 0). Therefore, $A = \{5, 7, 9\}$. The possibilities for y will be the hundreds digits of the products (3,305)(5) = 16,525, (3,305)(7) = 23,135, and (3,305)(9) = 29,745. Thus, y can be 5, 1, or 7, and so B = {1, 5, 7}.

The correct answer is D.

•

If x and y are integers such that $2 < x \leq 8$ and $2 < y \leq 9$, what is the maximum value of $\frac{1}{x} - \frac{x}{y}$?

A. $-3\frac{1}{8}$

B. 0

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C. $\frac{1}{4}$

D. $\frac{5}{18}$

E. 2

Algebra Inequalities

Because x and y are both positive, the maximum value of $\frac{1}{x} - \frac{x}{y}$ will occur when the value of $\frac{1}{x}$ is maximum and the value of $\frac{x}{y}$ is minimum. The value of $\frac{1}{x}$ is maximum when the value of x is minimum or when $x = 3$. The value of $\frac{x}{y}$ is minimum when the value of x is minimum (or when $x = 3$) and the value of y is maximum (or when $y = 9$). Thus, the maximum value of $\frac{1}{x} - \frac{x}{y}$ is $\frac{1}{3} - \frac{3}{9} = 0$.

The correct answer is B.

•

Items that are purchased together at a certain discount store are priced at $3 for the first item purchased and $1 for each additional item purchased. What is the maximum number of items that could be purchased together for a total price that is less than $30?

A. 25

B. 26

C. 27

D. 28

E. 29

Arithmetic Applied Problems

After the first item is purchased, $29.99 – $3.00 = $26.99 remains to purchase the additional items. Since the price for each of the additional items is $1.00, a maximum of 26 additional items could be purchased. Therefore, a maximum of 1 + 26 = 27 items could be purchased for less than $30.00.

The correct answer is C.

•

What is the least integer z for which $(0.000125)(0.0025)(0.00000125) \times 10^z$ is an integer?

A. 18

B. 10

C. 0

D. −10

E. −18

Arithmetic Decimals

Considering each of the three decimal numbers in parentheses separately, we know that 0.000125×10^6 is the integer 125, 0.0025×10^4 is the integer 25, and 0.00000125×10^8 is the integer 125. We thus know that $(0.000125) \times 10^6 \times (0.0025) \times 10^4 \times (0.00000125) \times 10^8 = (0.000125)(0.0025)(0.00000125) \times 10^6 \times 10^4 \times 10^8 = (0.000125)(0.0025)(0.00000125) \times 10^{6+4+8} = (0.000125)(0.0025)(0.00000125) \times 10^{18}$ is the integer $125 \times 25 \times 125$. We therefore know that if $z = 18$, then $(0.000125)(0.0025)(0.00000125) \times 10^z$ is an integer.

Now, if the product $125 \times 25 \times 125$ were divisible by 10, then for at least one integer z less than 18, $(0.000125)(0.0025)(0.00000125) \times 10^z$ would be an integer. However, each of the three numbers being

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multiplied in the product 125 × 25 × 125 is odd (not divisible by 2). We thus know that 125 × 25 × 125 is not divisible by 2 and is therefore odd. Because only even numbers are divisible by 10, we know that 125 × 25 × 125 is not divisible by 10. We thus know that 18 is the *least* integer z such that $(0.000125)(0.0025)(0.00000125) \times 10^z$ is an integer.

Note that it is not necessary to perform the multiplication 125 × 25 × 125.

The correct answer is A.

•

The average (arithmetic mean) length per film for a group of 21 films is t minutes. If a film that runs for 66 minutes is removed from the group and replaced by one that runs for 52 minutes, what is the average length per film, in minutes, for the new group of films, in terms of t?

A. $t + \frac{2}{3}$

B. $t - \frac{2}{3}$

C. $21t + 14$

D. $t + \frac{3}{2}$

E. $t - \frac{3}{2}$

Arithmetic Statistics

Let S denote the sum of the lengths, in minutes, of the 21 films in the original group. Since the average length is t minutes, it follows that $\frac{S}{21} = t$. If a 66-minute film is replaced by a 52-minute film, then the sum of the lengths of the 21 films in the resulting group is $S - 66 + 52 = S - 14$. Therefore, the average length of the resulting 21 films is $\frac{S-14}{21} = \frac{S}{21} - \frac{14}{21} = t - \frac{2}{3}$.

The correct answer is B.

•

A garden center sells a certain grass seed in 5-pound bags at $13.85 per bag, 10-pound bags at $20.43 per bag, and 25-pound bags at $32.25 per bag. If a customer is to buy at least 65 pounds of the grass seed, but no more than 80 pounds, what is the least possible cost of the grass seed that the customer will buy?

A. $94.03

B. $96.75

C. $98.78

D. $102.07

E. $105.36

Arithmetic Applied Problems

Let x represent the amount of grass seed, in pounds, the customer is to buy. It follows that $65 \le x \le 80$. Since the grass seed is available in only 5-pound, 10-pound, and 25-pound bags, then the customer must buy either 65, 70, 75, or 80 pounds of grass seed. Because the seed is more expensive per pound for smaller bags, the customer should minimize the number of the smaller bags and maximize the number of 25-pound bags to incur the least possible cost for the grass seed. The possible purchases are given in the table below.

x	Number of 25-pound bags	Number of 10-pound bags	Number of 5-pound bags	Total cost

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65	2	1	1	$98.78
70	2	2	0	$105.36
75	3	0	0	$96.75
80	3	0	1	$110.60

The least possible cost is then 3($32.25) = $96.75.

The correct answer is B.

•

If $x = -|w|$, which of the following must be true?

A. $x = -w$

B. $x = w$

C. $x^2 = w$

D. $x^2 = w^2$

E. $x^3 = w^3$

Algebra Absolute Value

Squaring both sides of $x = -|w|$ gives $x^2 = (-|w|)^2$, or $x^2 = |w|^2 = w^2$.

Alternatively, if (x, w) is equal to either of the pairs (−1, 1) or (−1, −1), then $x = -|w|$ is true. However, each of the answer choices except $x^2 = w^2$ is false for at least one of these two pairs.

The correct answer is D.

•

A certain financial institution reported that its assets totaled $2,377,366.30 on a certain day. Of this amount, $31,724.54 was held in cash. Approximately what percent of the reported assets was held in cash on that day?

A. 0.00013%

B. 0.0013%

C. 0.013%

D. 0.13%

E. 1.3%

Arithmetic Percents; Estimation

The requested percent can be estimated by converting the values into scientific notation.

$$\frac{31{,}724.54}{2{,}377{,}366.30} \quad \text{value as fraction}$$

$$= \frac{3.172454 \times 10^4}{2.37736630 \times 10^6} \quad \text{convert to scientific notation}$$

$$= \frac{3.172454}{2.37736630} \times \frac{10^4}{10^6} \quad \text{arithmetic property of fractions}$$

$$= \frac{3.172454}{} \times 10^{-2} \quad \text{subtract exponents}$$

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$= \frac{}{2.37736630} \times 10^{-4}$ subtract exponents

$\approx \frac{3}{2} \times 10^{-2}$ approximate

$= 1.5 \times 10^{-2}$ convert to decimal fraction

$= 0.015$ multiply

$= 1.5\%$ convert to percent

A more detailed computation would show that 1.3% is a better approximation. However, in order to select the best value from the values given as answer choices, the above computation is sufficient.

The correct answer is E.

$$\begin{array}{r} AB \\ +\,BA \\ \hline AAC \end{array}$$

•

In the correctly worked addition problem shown, where the sum of the two-digit positive integers AB and BA is the three-digit integer AAC, and A, B, and C are different digits, what is the units digit of the integer AAC?

A. 9

B. 6

C. 3

D. 2

E. 0

Arithmetic Place Value

Determine the value of C.

It is given that $(10A + B) + (10B + A) = 100A + 10A + C$ or $11A + 11B = 110A + C$. Thus, $11B - 99A = C$, or $11(B - 9A) = C$. Therefore, C is divisible by 11, and 0 is the only digit that is divisible by 11.

The correct answer is E.

•

The hard drive, monitor, and printer for a certain desktop computer system cost a total of $2,500. The cost of the printer and monitor together is equal to $\frac{2}{3}$ of the cost of the hard drive. If the cost of the printer is $100 more than the cost of the monitor, what is the cost of the printer?

A. $800

B. $600

C. $550

D. $500

E. $350

Algebra Simultaneous Equations

Letting d, m, and p, respectively, represent the cost of the hard drive, monitor, and printer, the following equations are given:

1. $d + m + p = 2{,}500$

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1. $d + m + p = 2{,}500$
2. $p + m = \frac{2}{3}d$
3. $p = m + 100$

Using (2) and substituting $\frac{2}{3}d$ for $m + p$ in (1) gives $\frac{5}{3}d = 2{,}500$, from which $d = 1{,}500$. Then from (1), $p + m = 1{,}000$, but $m = p - 100$ from (2), so $2p - 100 = 1{,}000$, $2p = 1{,}100$, and $p = 550$.

The correct answer is C.

$$3r \leq 4s + 5$$
$$|s| \leq 5$$

•

Given the inequalities above, which of the following CANNOT be the value of r?

A. −20
B. −5
C. 0
D. 5
E. 20

Algebra Inequalities

Since $|s| \leq 5$, it follows that $-5 \leq s \leq 5$. Therefore, $-20 \leq 4s \leq 20$, and hence $-15 \leq 4s + 5 \leq 25$. Since $3r \leq 4s + 5$ (given) and $4s + 5 \leq 25$ (end of previous sentence), it follows that $3r \leq 25$. Among the answer choices, $3r \leq 25$ is false only for $r = 20$.

The correct answer is E.

•

If m is an even integer, v is an odd integer, and $m > v > 0$, which of the following represents the number of even integers less than m and greater than v?

A. $\frac{m-v}{2} - 1$
B. $\frac{m-v-1}{2}$
C. $\frac{m-v}{2}$
D. $m - v - 1$
E. $m - v$

Arithmetic Properties of Numbers

Since there is only one correct answer, one method of solving the problem is to choose values for m and v and determine which of the expressions gives the correct number for these values. For example, if $m = 6$ and $v = 1$, then there are 2 even integers less than 6 and greater than 1, namely the even integers 2 and 4. As the table below shows, $\frac{m-v-1}{2}$ is the only expression given that equals 2.

$$\frac{m-v}{2} - 1 = 1.5$$
$$\frac{m-v-1}{2} = 2$$

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$$\begin{aligned} \frac{m-v}{2} &= 2.5 \\ m-v-1 &= 4 \\ m-v &= 5 \end{aligned}$$

To solve this problem it is not necessary to show that $\frac{m-v-1}{2}$ always gives the correct number of even integers. However, one way this can be done is by the following method, first shown for a specific example and then shown in general. For the specific example, suppose $v = 15$ and $m = 144$. Then a list—call it the first list—of the even integers greater than v and less than m is 16, 18, 20, …, 140, 142. Now subtract 14 (chosen so that the second list will begin with 2) from each of the integers in the first list to form a second list, which has the same number of integers as the first list: 2, 4, 6, …, 128. Finally, divide each of the integers in the second list (all of which are even) by 2 to form a third list, which also has the same number of integers as the first list: 1, 2, 3, …, 64. Since the number of integers in the third list is 64, it follows that the number of integers in the first list is 64. For the general situation, the first list is the following list of even integers: $v+1, v+3, v+5, \ldots, m-4, m-2$. Now subtract the even integer $v-1$ from (i.e., add $-v+1$ to) each of the integers in the first list to obtain the second list: 2, 4, 6, …, $m-v-3, m-v-1$. (Note, for example, that $m-4-(v-1) = m-v-3$.) Finally, divide each of the integers (all of which are even) in the second list by 2 to obtain the third list: 1, 2, 3, …, $\frac{m-v-3}{2}, \frac{m-v-1}{2}$.

Since the number of integers in the third list is $\frac{m-v-1}{2}$, it follows that the number of integers in the first list is $\frac{m-v-1}{2}$.

The correct answer is B.

•

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